Kanav Dutta

+1 (587) 545-5631 kdutta1@ualberta.ca linkedin.com/in/kanavdutta github.com/kanavdutta

Summary

Computing Science student at the University of Alberta and current research assistant with hands-on AI tooling, agentic systems, and Azure deployment experience. Built local AI tools for multi-step tasks and improved a data processing pipeline to reduce turnaround time by 57%. Enrolled in a co-op/internship program and available January - December 2027.

Education

B.Sc. in Computing Science Specialization Edmonton, AB
University of Alberta *2022 - 2027*
Areas of Study: Software Engineering, Web Applications & Architecture, Databases, Algorithms, Cloud Computing.

Technical Skills

Languages	Python, TypeScript, JavaScript, Go, SQL, HTML5, CSS3
Frontend	React.js, React Native, Next.js
Backend	Node.js, REST APIs, schema validation, async queues
AI & GenAI Tooling	Claude, Ollama, local embeddings/RAG, agentic tool-calling systems
DevOps & Tools	Azure, Git, GitHub, Docker, CI/CD, Unity, SteamVR
Databases	PostgreSQL, SQLite, MongoDB

Experience

Research Assistant Aug 2026 - Present
Memory for Events Lab, University of Alberta *Edmonton, AB*

- Helping develop a Unity script for use with SteamVR in a psychology research study examining participant reactions; using Claude to assist with ongoing script development.

Software Engineering Intern Sep 2022 - Sep 2023
Adobe Inc. *Edmonton, AB*

- Collaborated with development, QA, and project management stakeholders in an Agile team through daily standups and sprint cycles.
- Built and shipped full-stack TypeScript and JavaScript features, including React components, REST API integrations, and backend logic, supported by automated unit tests and peer code review.
- Diagnosed a data-fetching bug affecting state across 3 components; improved page performance by 15% and eliminated recurring user-facing errors.

Software Development Intern Sep 2021 - Mar 2022
Fore Solutions Pvt. Ltd. *New Delhi, India*

- Delivered full-stack features within sprint deadlines, taking ownership from design through testing and deployment.
- Re-architected a data processing pipeline, reducing turnaround time by 57% while maintaining output integrity, and documented the change for ongoing team reference.

Projects

Local AI Agent Toolkit *Python, Ollama, SQLite, Local Embeddings, Agentic AI*

- Built fully local AI tools, including an assistant with SQLite-backed memory, local embeddings, and a tool-calling agent loop that plans and executes multi-step tasks.

AssistAI *TypeScript, React Native, Node.js, REST APIs, PostgreSQL, Azure, JWT Auth*

- Built a full-stack application with REST API endpoints, JWT authentication, and structured request/response validation.
- Deployed backend services to Azure through a CI/CD pipeline that ran the full unit test suite on every commit.

SignalFence *TypeScript, Node.js, REST API, PostgreSQL, Docker, CI/CD, Unit Testing*

- Developed a REST API service with PostgreSQL persistence and Docker deployment, integrating automated regression tests into CI/CD.

Awards

- University of Alberta Regional Excellence Scholarship
- University of Alberta International Admission Scholarship